TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Digital Twin Framework for Manufacturing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( )			
Industrial Technology Classification [1]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Machinery and materials		Machinery and materials	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial/ General Machinery	Industrial/General Machinery Software	Component & Element Parts	Component-related Software
Concept of Proposed Technology	This technology proposal introduces a **digital twin framework that enables manufacturers to create and utilize unified, interoperable digital representations of their machines, production systems, and processes,** providing a modern foundation for smarter, more efficient, and more sustainable manufacturing [1][2]. The framework consolidates data from heterogeneous machines, systems, and workflows into a coherent digital ecosystem, replacing fragmented information silos with integrated and continuously updated data structures [6]. The way how operational data is collected, structured, and interpreted, ensures that digital representations accurately reflect the real-world behavior of assets across the production environment [7]. In a manufacturing context, an asset refers to any physical equipment, machine, tool, production line, or infrastructure component that plays a functional role in the production process and whose operational state, condition, performance, or quality-related			

	information can be digitally represented and monitored. This digital backbone is built upon the international standard IEC 63278, which defines the structure of the Asset Administration Shell (AAS), a globally recognized model for standardized digital representations of industrial assets [1][6]. Because this foundation follows a widely adopted standard, interoperability can be extended across the entire value chain, while digital twin descriptions remain unified, standardized, and machine-readable [6][7]. This enables automation not only within a manufacturing company but also across interconnected supply chains, e.g. demonstrated in Catena-X [8]. Another beneficiary feature of the technology is its scalable architecture, which allows manufacturers to start with individual assets and extend the digital twin layer across entire production lines, factories, and value-chain networks without re-engineering core components [2][12]. Moreover, it is designed to be applicable across any manufacturing sector and remains fully agnostic to specific industrial branches, ensuring broad usability beyond traditional production environments [7]. In addition, it leverages open-source technologies such to promote transparency, extensibility, and collaborative innovation [11][12]. By establishing a reliable and semantically consistent data layer, the digital twin framework effectively prepares manufacturing environments for AI-enabled operations. Since AI methods depend on high-quality, context-rich, and well-structured data, their effectiveness is often limited by inconsistent or incomplete information sources. By offering a uniform and standardized representation of assets, digital twins ensure that future AI systems can access validated, contextualized, and operationally relevant data [2][14]. This capability enables manufacturers to act more proactively, reduce operational disruptions, and continuously improve performance across the full production lifecycle once AI tools are applied. The core benefit behind the technology is that it provides manufacturers with an accessible and structured method to develop digital twins that aggregate asset data into consistent digital information models, which together constitute a standardized description of each asset [1][6].The framework provides tools to model all types of machinery and workflows and supports both static and dynamic implementations for handling data—for example, to track behavior over time or to manage component states—ensuring that digital twins remain accurate, maintainable, and continuously aligned with real-world operations [7][11].
Rationale for Joint R&D Project	The proposed joint R&D project is essential to successfully develop, validate, and industrialize the digital twin–based technologies described in this proposal. Collaboration is required for several reasons: **1. Create innovation space:** The combination of German long-term expertise in digital-twin modeling, European industrial data standards, and the advanced manufacturing knowledge and innovative engineering practices of Korean SMEs provides a strong foundation for developing new and innovative solutions [6]. **2. Build a shared, future-oriented vision for digital twins:** Korea's highly connected manufacturing systems offer use cases that challenge current European technologies and demand new approaches. This allows both partners to explore new architectural ideas and develop forward-looking concepts,

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	for instance, integrating digital twins with AI-technology [15]. Instead of focusing solely on development, the project helps establish a joint understanding of how digital twins, data semantics, and AI should evolve for next-generation manufacturing. **3. Learn from Korean industry:** Korean SMEs bring experience from environments where decisions are made quickly and supply chains are closely aligned, offering valuable insights for developing more responsive and interconnected digital-twin solutions. **4. Combine practical use cases with standardized concepts** Korean SMEs gain access to semantic modeling methods, interoperability concepts, and experience with internationally aligned standards (such as the AAS). By merging Korean real-world scenarios with our methodological foundations, the project delivers solutions that are conceptually robust and validated under real industrial conditions. **5. Deliver globally relevant and deployable results** Bringing together the methodology, practices, and constraints of two different industrial cultures creates outcomes that are more adaptable, better validated, and more suitable for international markets [1][6]. Together, these factors demonstrate the necessity of a joint R&D effort and highlight the strategic relevance of collaboration under the 2026 TIPA International Joint R&D Program.
Key Words	1.Digital Twin 2. Interoperability 3. Asset Administration Shell (IEC 63278-Series) 4. Manufacturing Data Integration 5. AI-Ready Industrial Data Layer

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level ( )	End of the Project	Level ( )
R&D Objectives	The project has following key objectives: **1. Development of a Standardized Digital-Twin Framework** The first objective is to develop a standardized digital-twin framework for manufacturing assets and focusing on one manufacturing use case, created in close cooperation with a Korean SME, to ensure that machines, production lines, and processes can be represented in a unified and interoperable manner suitable for diverse industrial ecosystems. **2. Analysis and Review of a Digital-Twin-Ready IT-Architecture** The second objective is to establish an IT architecture concept that reliably connects enterprise IT systems with the digital twin framework, enabling consistent data collection, processing, and transfer from heterogeneous machines and production systems so that digital twins accurately reflect the real-time operational state of assets. **3. Development of a Digital Twin Methodology for Korena SME's** The third objective is to develop a comprehensive methodology that enables Korean SMEs to adopt a practical set of industrial tools, including modeling capabilities, standardized templates, and foundational software components. This methodology should rely on open-source technologies to promote transparency, adaptability, and broad industrial uptake. **4. Validation and Demonstration of the Digital-Twin Framework** The fourth objective is to validate the developed digital-twin framework and associated methodology within real Korean SME manufacturing environments, combining European methodological expertise with Korean operational practices to ensure practical applicability, scalability, and industrial relevance. **5. Dissemination nd Exploration of Future-Oriented Concepts for AI-Ready Digital Twins** The fifth objective is to explore future-oriented concepts that prepare digital twins for AI integration, by ensuring that manufacturing data is structured, semantically enriched, and of sufficient quality to support advanced analytics, predictive applications, and emerging AI-driven manufacturing scenarios.			
R&D Target Specifications	The target performance specifications of this R&D project are defined around o**ne, or at most three, manufacturing use cases** provided by the Korean SME partner. The overarching goal is to demonstrate that the developed digital-twin framework, the associated IT-architecture concept, and the accompanying methodological guidance can be successfully applied to real operational scenarios and evaluated against clearly defined requirements. Thus, the first target of the project is to conduct **a digital-twin-ready IT-architecture review.** The review document should provide insights on how enterprise IT systems, operational data sources, and data-integration components interact to support the selected use case(s). This specification will document all relevant system interfaces, data flows, integration pathways, and software components in a structured and transferable form, enabling subsequent industrial implementation.			

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	Another target of the project is the creation of **a practical digital-twin methodology tailored for Korean SMEs**. This methodology will provide clear guidance on how to model and manage digital-twin structures, how to apply standardized elements and submodels aligned with international standards, and how to adapt the architectural concepts to additional assets or processes beyond the pilot use case(s). The methodology will also include a concise summary of **at least eight practical lessons learned**, covering aspects such as reference model design, construction of system diagrams, documentation of component interactions, and recommended practices for SME adoption. The final target of the project is the **validation and demonstration of the selected use case(s)** in cooperation with the Korean SME partner. This will be carried out within a dedicated demonstration environment, showcasing the capabilities of the digital-twin framework and its core functionalities under realistic manufacturing conditions. The demonstration will confirm that the developed IT-architecture concept and digital-twin framework can support the intended functionalities and are appropriate for practical deployment in SME-scale manufacturing environments.
R&D Plan	********** PHASE 1 ********** **WP1 – Use-Case Definition and Requirements Analysis** Objective: Define the technical and operational requirements based on up to three manufacturing use cases. **Task 1:** Use-Case Selection and Documentation **Task 2:** Asset, Data Source, and IT System Analysis **Task 3:** Requirements Analysis and Consolidation Deliverables: Use Case and Requirements Report Milestone: M1 – Requirements approved **WP2 –Digital-Twin-Ready IT-Architecture Review** Objective: Complete the IT architecture review supporting digital-twin creation and data integration. **Task 1:** Initial IT-Architecture Review and Requirements Analysis **Task 2:** Digital Twin Interface Definition and Data Flow Description **Task 3:** Final Report on the Digital-Twin-Ready IT-Architecture Review Deliverables: IT-Architecture Review Documentation Milestone: M2 – IT Architecture Concept finalized **WP3 – Development of the Digital Twin Methodology for SMEs** Objective: Provide a practical methodology enabling SMEs to adopt digital-twin concepts. Task 1: Development of the Modeling Guideline Task 2: Adaptation to SME Operational Conditions Task 3: Dissemination Plan) and Lessons Learned (Light Dissemination) Deliverables: SME Methodology Guide, Templates, Lessons-Learned Summary Milestone: M3 – SME Methodology completed

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********** PHASE 2**********

WP4 – Development of the Standardized Digital Twin Framework

Objective: Create the conceptual structure and standardized modeling approach for the digital twin.

Task 1: Digital Twin Modeling for Selected Use Case(s)

Task 2: Framework Set-Up

Task 3: Initial Testing

Deliverables: Digital Twin Framework with the Set of Information Models

Milestone: M3 – Digital Twin Framework Concept completed

WP5 – Use Case Implementation and Proof-of-Concept

Objective: Implement the selected use case(s) and perform the Proof of Concept to validate the digital-twin framework and IT-architecture.

Task 1: Implementation of Digital Twin Models for the Selected Use Case(s)

Task 2: Data Integration, Connectivity, and System Configuration

Task 3: Execution of the Proof-of-Concept and Validation

Deliverables: Use Case Implementation and PoC Report

Milestone: M5 – PoC successfully completed

WP6 – Demonstration

Objective: Implement a demonstration showcasing selected functionalities of the digital twin use case(s).

Task 1: Development of the Demonstration Scenario

Task 2: Demonstration-Scenario Implementation

Task 3: Internal Testing and Refinement

Deliverables: Demonstration Environment, Scenario Documentation

Milestone: M5 – Demo environment operational

WP7 – Project Management, Dissemination and Exploitation

Objective: Manage the project, communicate project results and ensure transfer of knowledge to stakeholders, industry, and the research community in Korea and Germany.

Task 1: Project Management

Task 2: Dissemination

Task 3: Standardization and Exploitation

Deliverables: Dissemination Package, Communication Materials, Final Dissemination Report; Standardization Meetings

Milestone: M7 – Dissemination completed

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Research Institute will provide leadership in the development of the IT architecture, the digital-twin methodology, and the standardized digital-twin framework. It will lead the use-case analysis, guide the conceptual design of the technical solution, and take responsibility for preparing and setting up the system environment required for proof-of-concept of technologies. The Institute will also support the implementation of the demonstration, ensure the technical coherence of all components, and contribute to dissemination activities, including workshops and joint events, if any. In addition, the Institute will lead and contribute to standardization efforts, including German and joint Korean–German standardization activities, to promote alignment with international standards and digital twin vision.
Expected R&R of Korean SME Partner	The Korean SME Partner will support the project through active use-case development, providing all necessary requirements, operational insights, and technical documentation. It will implement the required use-case APIs, system interfaces, tools, and services needed for integration with the digital-twin framework and the use case(s). The SME will also support or conduct the demonstration in the SME environment, ensure that the developed components function under realistic conditions, and participate in dissemination, exploitation, and—if possible—standardization activities. SME acts as the primary industrial testbed to ensure practical relevance and applicability.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Results:** The joint R&D project is expected to deliver a set of concrete results that demonstrate the technological feasibility, industrial applicability, and practical value of a standardized digital-twin approach. The project will produce an interoperable digital-twin framework based on international standard such as IEC 63278, including information models, submodels, and reusable templates that support consistent and machine-readable digital representations of assets. In addition, a complete digital-twin-ready IT architecture will be designed and validated, showing how heterogeneous machines, enterprise IT systems, and digital-twin services can be connected in a unified, scalable, and interoperable manner. A practical methodology tailored for SMEs will be developed, providing structured modeling guidelines, reference structures, templates, tools and lessons learned to help smaller companies adopt digital-twin technologies effectively and cost-efficiently. As part of the use case implementation, the partners will develop and validate a fully functioning proof-of-concept that demonstrates end-to-end data integration and confirms the technical soundness of the framework under real operational conditions. A demonstration environment will be created to showcase the capabilities of the digital twin, providing a transferable model that can be applied across manufacturing sectors and extended to additional assets or production processes. Finally, the project will produce significant contributions to knowledge transfer and standardization. Documentation, insights, and technical recommendations will support ongoing German–Korean standardization activities and strengthen alignment with emerging international practices. The results will therefore be relevant not only for the participating organizations but for broader industrial ecosystems seeking interoperable and scalable digital-twin solutions. **Intended Impact:** The intended impact of the project extends well beyond the immediate technical outputs, fostering long-term industrial and collaborative benefits. By enabling SMEs to apply practical and standardized digital-twin solutions, the project will strengthen industrial competitiveness through improved transparency and efficiency. The availability of a consistent and semantically rich data backbone will facilitate the adoption of AI-enabled applications, predictive analytics, and automation, accelerating digital transformation across manufacturing sectors and supporting broader Industry 4.0 adoption. The project outcomes offer strong commercialization potential by enabling consulting services, integration offerings, training programs, and reusable software components. The framework's modular design allows manufacturers to start with individual assets and extend digital-twin capabilities across entire production lines or supply chains without re-engineering their systems. Through joint development, the project will also reinforce Korean–German collaboration, combining European expertise in standardized digital-twin modeling. This cooperation will help harmonize digital-twin practices across both countries and contribute to evolving international standards, improving interoperability and easing cross-border deployment. Over the long term, both partners will gain reusable technical assets, methodological foundations, and operational experience that support continued innovation, further R&D initiatives, and deployment in future industrial environments.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

Advanced Robotics & Manufacturing

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

Phase I (Pre-Study): TRL 2–3

Phase II (Main R&D): TRL 4–6